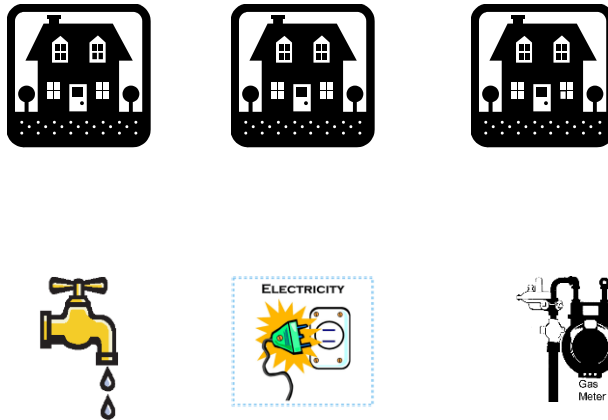


Planar Graphs

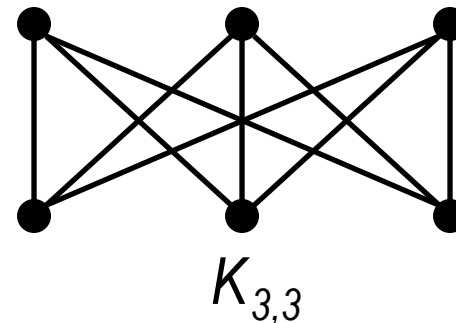
Planar Graphs

● *The Three Utilities and Three Houses puzzle:*

Is it possible to join three houses to each of three separate utilities so that none of the connections cross?

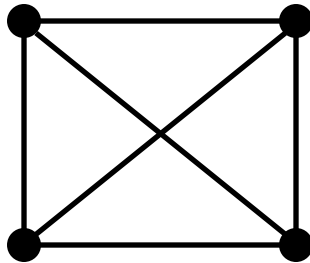
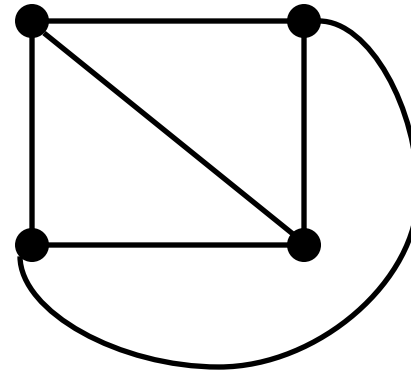


● *This question can be rephrased as: Can the complete bipartite graph $K_{3,3}$ be drawn on the plane such that no two of its edges cross?*



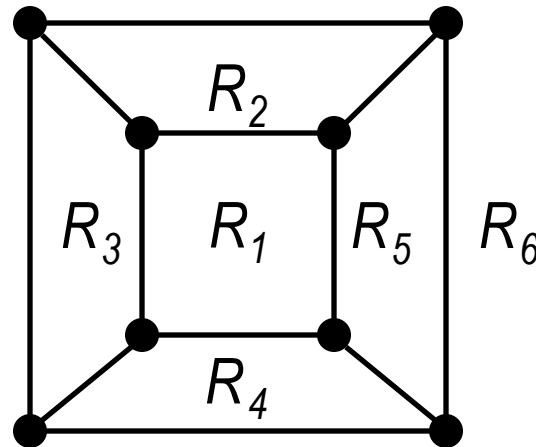
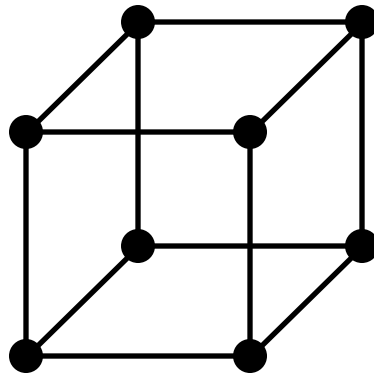
Planar Graphs (cntd)

- A graph is called **planar** if it can be drawn in the plane without any edges crossing (where a crossing of edges is the intersection of lines or arcs representing them at a point other than their common endpoint). Such a drawing is called a **planar representation** of the graph

 K_4 *planar representation of K_4*

Euler's Formula

- *A planar representation of a graph splits the plane into regions, including the unbounded region.*

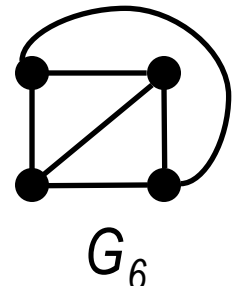
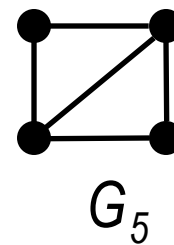
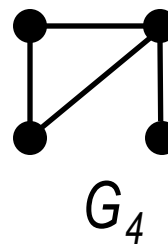
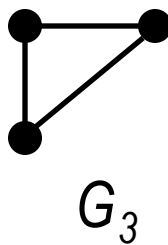
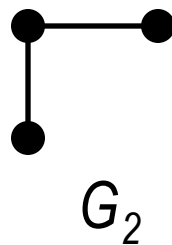


- **Euler's Formula.**

Let G be a connected planar simple graph with e edges and v vertices. Let r be the number of regions in a planar representation of G . Then $v - e + r = 2$.

Euler's Formula (cntd)

- *We prove the formula by structural induction.*
- *Fix a planar representation of G .*
- *We construct a sequence of graphs $G_1, G_2, \dots, G_e = G$ as follows:*
 - *Arbitrary pick an edge of G to obtain G_1*
 - *Obtain G_n from G_{n-1} by arbitrary adding an edge of G that is incident with a vertex already in G_{n-1} . This construction is possible because G is connected.*
- *G is obtained after e edges are added*
- *Let r_n , e_n , and v_n represent the number of regions, edges, and vertices of G_n*



Euler's Formula (cntd)

- *Basis step.*

For G_1 we have $r_1 = e_1 = 1$ and $v_1 = 2$ and so $v_1 - e_1 + r_1 = 2$

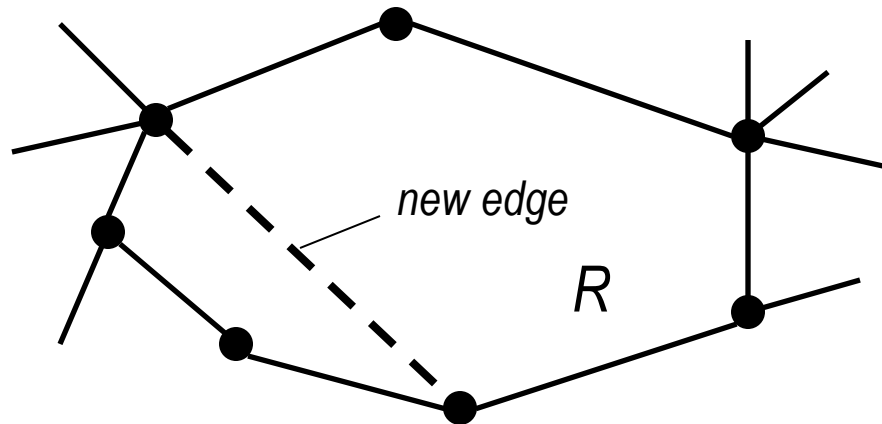
- *Inductive step.*

Suppose that $v_k - e_k + r_k = 2$. We will prove $v_{k+1} - e_{k+1} + r_{k+1} = 2$

- *We consider two cases.*

Euler's Formula (cntd)

- *Case 1. The new edge connect two vertices that are present in G_k . These two vertices are on the border of the same region because we use a planar representation of G .*

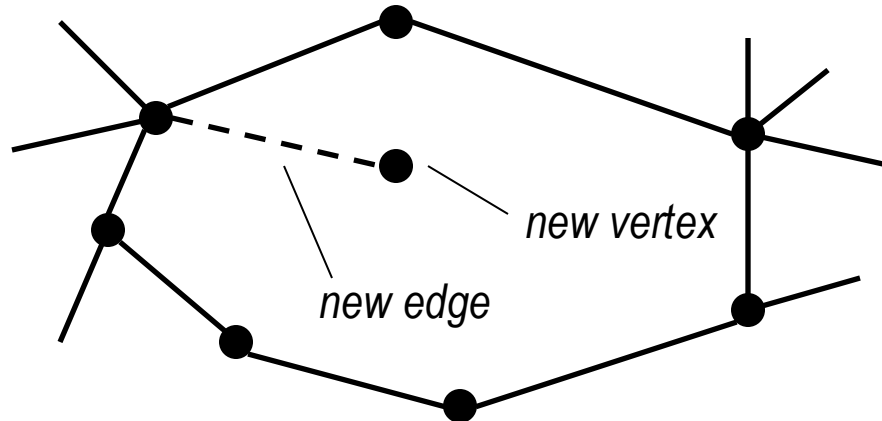


- *The new edge splits this region into two. Therefore $r_{k+1} = r_k + 1$, $e_{k+1} = e_k + 1$ and $v_{k+1} = v_k$. Thus,*

$$v_{k+1} - e_{k+1} + r_{k+1} = v_k - (e_k + 1) + (r_k + 1) = v_k - e_k + r_k = 2$$

Euler's Formula (cntd)

- *Case 2. The new edge is pendant, and therefore a new vertex is added.*



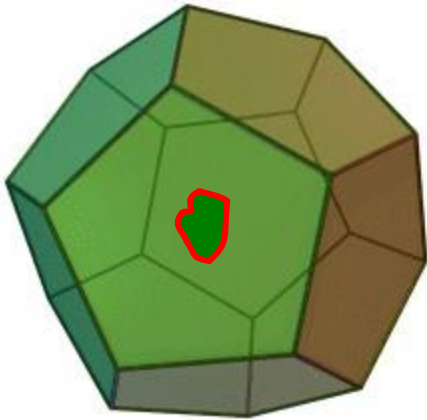
No new region is produced. Therefore $r_{k+1} = r_k$, $e_{k+1} = e_k + 1$, and $v_{k+1} = v_k + 1$. Thus,

$$v_{k+1} - e_{k+1} + r_{k+1} = (v_k + 1) - (e_k + 1) + r_k = v_k - e_k + r_k = 2$$

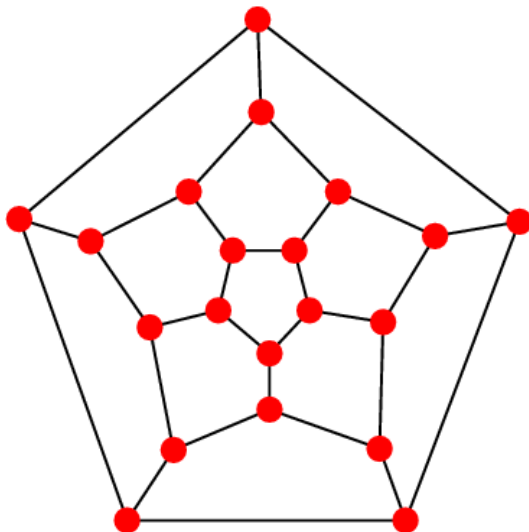
Q.E.D.

Polyhedrons and Planar Graphs

- *Every polyhedron can be converted into a planar graph*



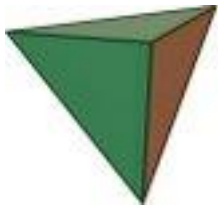
- *Make a hole in the rubber polyhedron
Expand it flat
Put it in the plane
We obtain a planar graph*



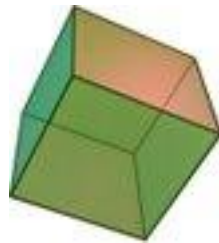
- *Vertices become vertices
Edges become edges
Faces become regions*

Platonic Solids

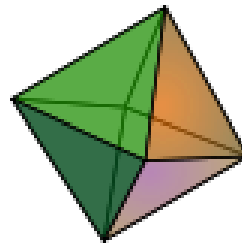
● *The simplest regular polytops*



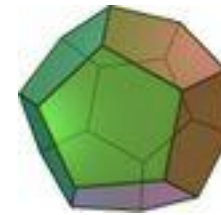
Tetrahedron



Hexahedron



Octahedron

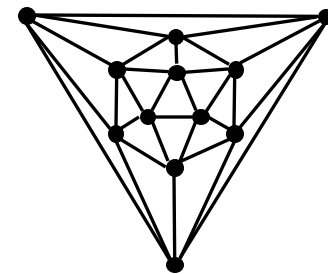
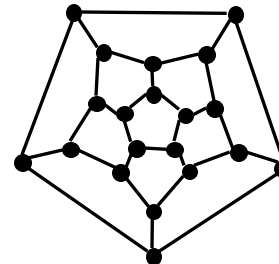
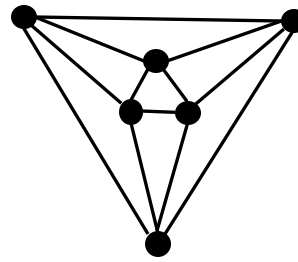
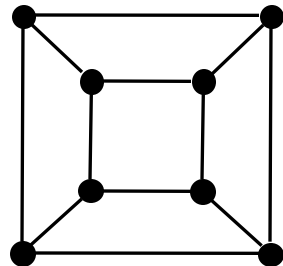
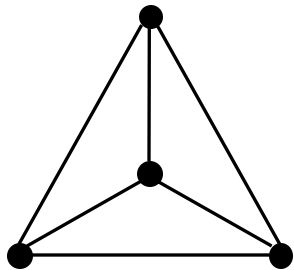


Dodecahedron



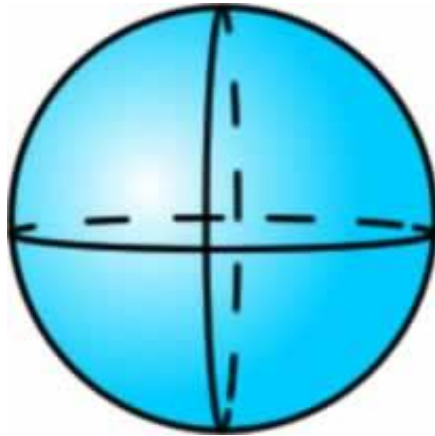
Icosahedron

● *The corresponding graphs*



Euler Characteristics

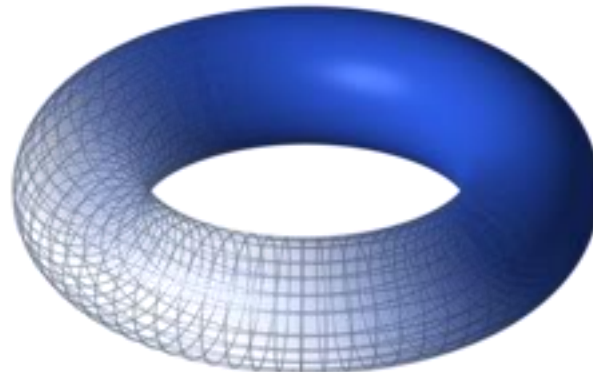
- Imagine again that a polyhedron is made of rubber. Instead of tearing it, inflate it.



- We obtain a sphere with the graph of the polyhedron drawn on it
- This graph is planar. Therefore we know that

$$v - e + r = 2$$

- 2 is the **Euler characteristic** of a sphere
- What if the polyhedron is such that after inflating we obtain a **torus**?



Non-Planar Graphs

● Corollary.

Let G be a connected planar graph with v vertices, e edges, and r regions. Then $e \leq 3v - 6$.

● Proof.

Clearly, the boundary of every region contains at least 3 edges. Let these numbers be $l_1, l_2, \dots, l_r$.

Go around every region. The total number of edges we use is

$$l_1 + l_2 + \dots + l_r \geq 3r.$$

*On the other hand, we have traveled along every edge at most twice
Therefore, $3r \leq 2e$. By Euler's formula*

$$3v - 3e + 3r = 6$$

$$3e - 3v + 6 = 3r \leq 2e$$

$$3e - 3v + 6 \leq 2e \Rightarrow e \leq 3v - 6$$

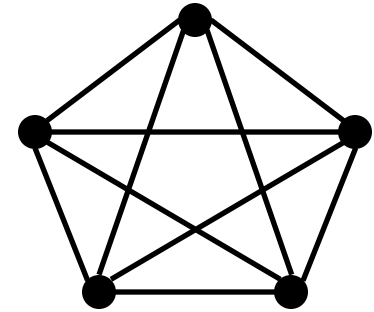
Non-Planar Graphs (cntd)

- The graph K_5 is not planar.
- Indeed, K_5 has $v = 5$ vertices and $e = 10$ edges.
$$e = 10 > 9 = 3v - 6$$
- Graph $K_{3,3}$ is not planar.
- Note that $K_{3,3}$ is bipartite. Therefore every cycle has length at least 4.

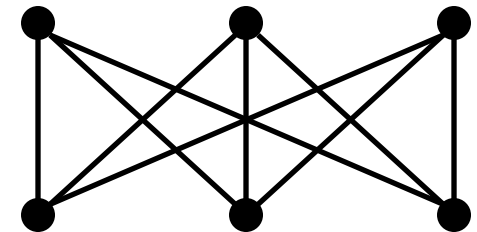
As in the corollary above we get the inequality $4r \leq 2e$

By Euler's formula, $r = e - v + 2 = 9 - 6 + 2 = 5$.

However, $4r = 20 > 18 = 2e$



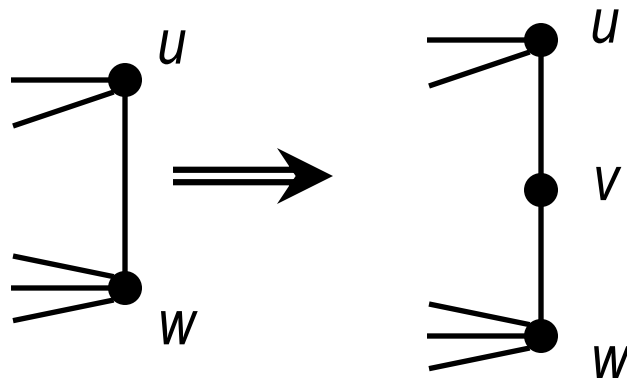
K_5



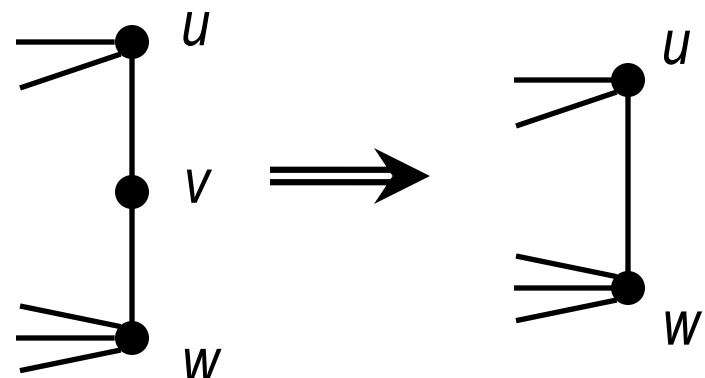
$K_{3,3}$

Subdivisions and Exclusions

- Let $G = (V, E)$ be a simple graph. An **elementary subdivision** of G results when an edge $e = uw$ is removed from G and then edges uv and vw are added to $G - e$, where $v \notin V$.
- Let $v \in V$ be a vertex of degree 2 in G , and let $uv, vw \in E$, where u and w are not adjacent. An **exclusion** of G results when the vertex v and edges uv and vw are removed from G and then the edge uw is added.



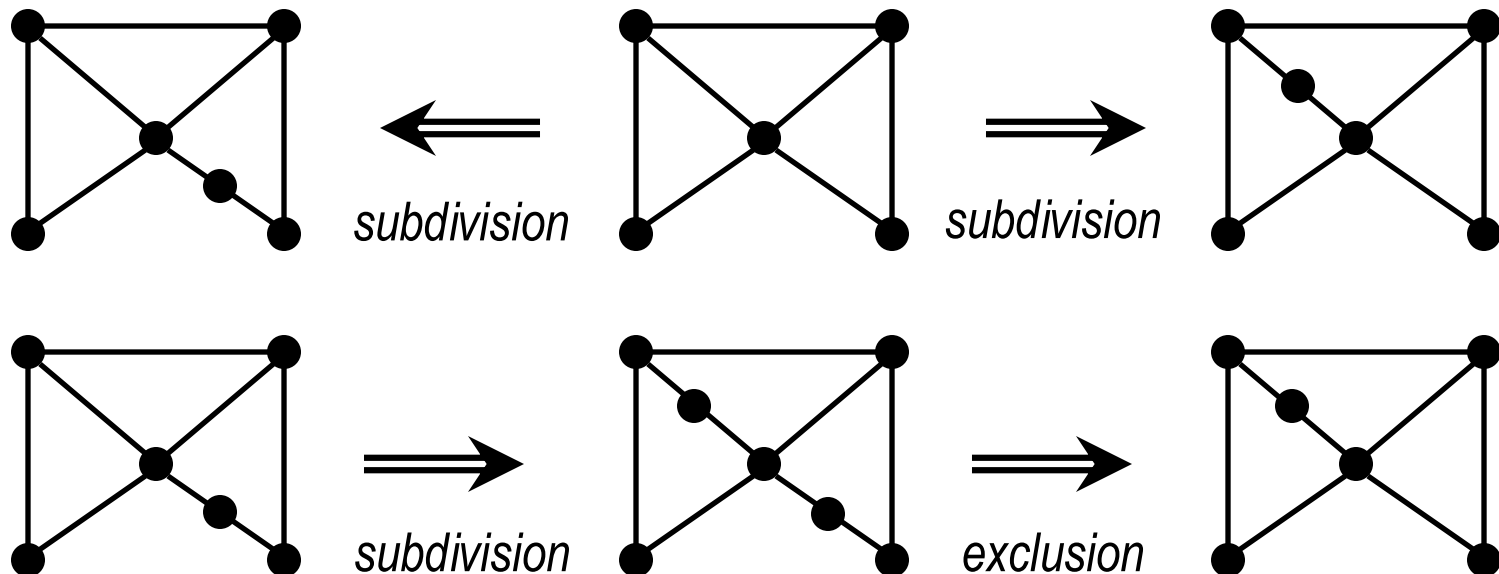
elementary subdivision



exclusion

Homeomorphic Graphs

- Simple graphs G and H are called homeomorphic if they are isomorphic or if they can both be obtained from the same simple graph by a sequence of elementary subdivisions
- Equivalently, the graphs G and H are called homeomorphic if they are isomorphic or if G can be obtained from H by a sequence of elementary subdivisions and exclusions.



Kuratowski's Theorem

● Kuratowski's Theorem.

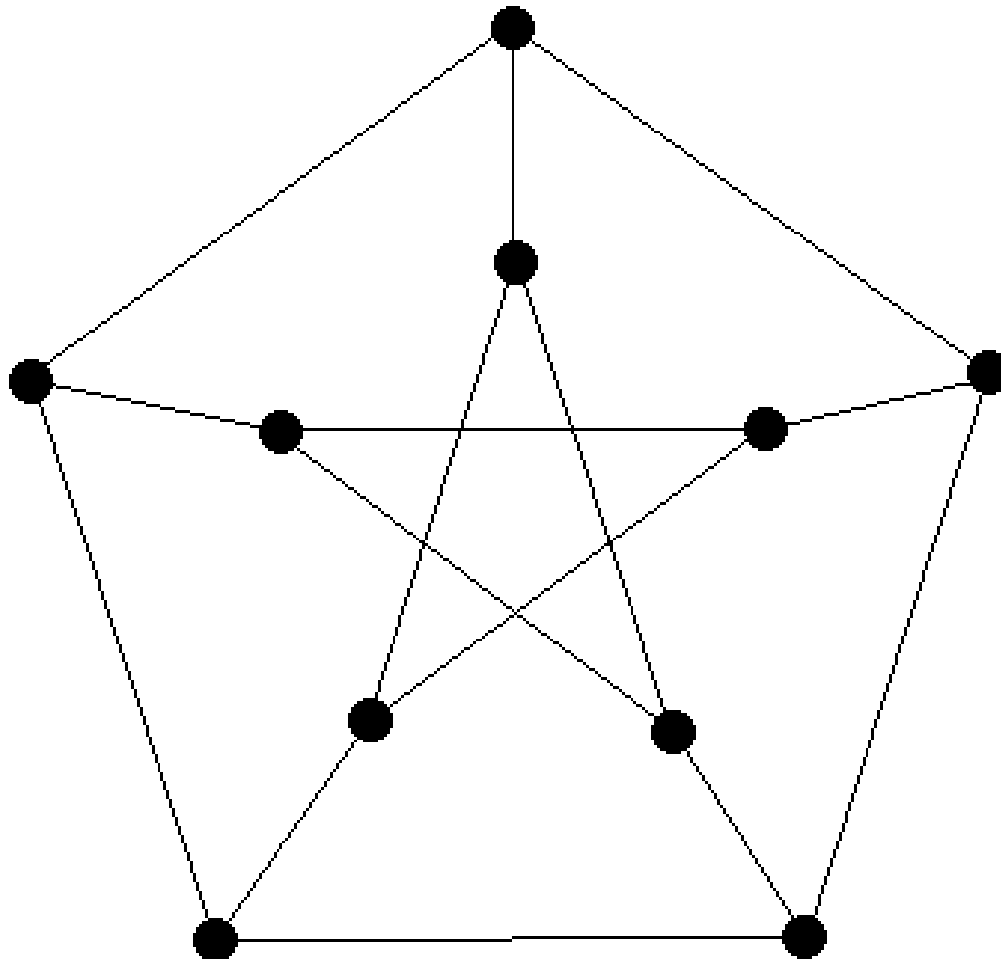
A graph is nonplanar if and only if it contains a subgraph that is homeomorphic to either K_5 or $K_{3,3}$

● Proof.

One direction is easy: If a subgraph is nonplanar then the entire graph is nonplanar. Also, elementary subdivisions and exclusions do not destroy planarity.

The other direction is hard.

Petersen Graph



Practice

Exercises from the Book:

7th edition: 1, 13, 19, 22 (page 725 – 726)

8th edition: 1, 13, 19, 22 (page 760 – 761)